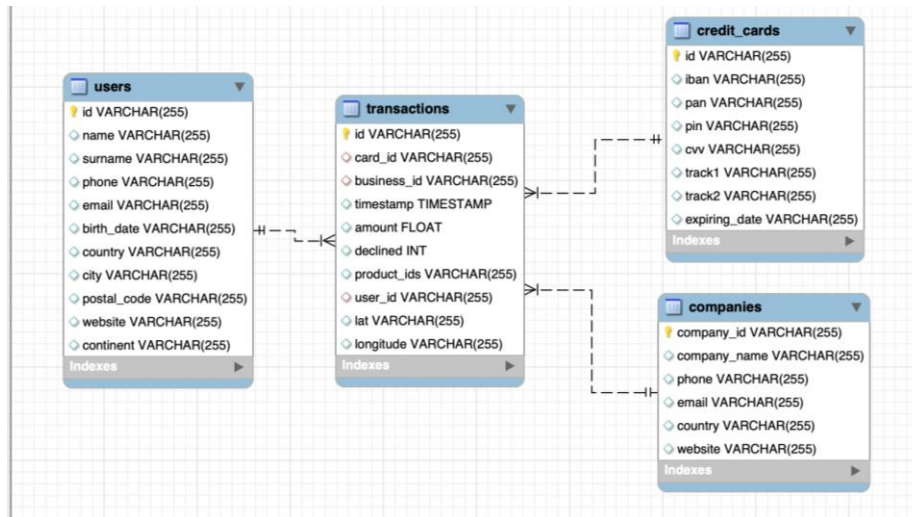


Nivell 1

Descàrrega els arxius CSV, estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguem realitzar les següents consultes:



Mostro la creació de la DB “sprint_04_transactions” amb model en estrella on la fact table es “transactions” i les “dimension” tables són “users”, “credit_cards” i “companies”.

- Taula credit_cards: PK = “id”, fem DROP amb “user_id” per eliminar relació amb la taula “users” i poder crear el model estrella.
- Taula “users”: PK=“id”, creem columna continent per unir els datasets d’ambdós continents.
- Taula “companies”: PK=“company_id”
- Taula “transactions”: PK=“id”, creem restricció tipus FK amb les taules de dimensions amb les columnes “card_id”, “business_id” i “user_id” i les respectives PK de cada taula.

Exercici 1

Realitza una subconsulta que mostri tots els usuaris amb més de 80 transaccions utilitzant almenys 2 taules.

```

123 • SELECT *
124 FROM users AS u
125 WHERE EXISTS(
126     SELECT user_id
127     FROM transactions AS t
128     WHERE u.id = user_id
129     GROUP BY user_id
130     HAVING COUNT(id) > 80
131 );

```

100% 15:127

Result Grid Filter Rows: Search Edit: Export/Import:

id	name	surname	phone	email	birth_date	country	city	postal_code
185	Molly	Gilliam	0800 120 8023	donec@outlook.couk	Dec 21, 1993	United Kingdom	London	EC1A 1BB
289	Dxwgi	Hwcru	+98-309-8797	dxwgi.hwcru@example.com	Aug 20, 1976	Germany	Stuttgart	70173
318	Bnyr	Astuw	+33-120-9644	bnyr.astuw@example.com	May 3, 1974	Italy	Genoa	16100
454	Sfzzoh	Xavfridxs	+58-495-3945	sfzzoh.xavfridxs@examole.com	Aug 28, 1962	Poland	Gdansk	80-001

users 9

Action Output

	Time	Action	Response
34	13:18:11	SELECT product_id, COUNT(product_id) AS sales_for_product FROM transact...	100 row(s) returned
35	10:19:37	SELECT * FROM users WHERE id IN(SELECT user_id FROM transaction...	4 row(s) returned
36	10:20:30	SELECT * FROM users AS u WHERE EXISTS(SELECT user_id FROM tra...	4 row(s) returned
37	10:28:50	SELECT * FROM sprint_04_transactions.transaction_products	253391 row(s) returned
38	10:32:31	SELECT product_id, COUNT(product_id) AS sales_for_product FROM transact...	100 row(s) returned
39	10:45:02	SELECT * FROM users AS u WHERE EXISTS(SELECT user_id FROM tra...	4 row(s) returned

Exercici 2

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```

140 /*Exercici 2
141 Mostra la mitjana d'amount per IBAN de les targetes de crèdit a
142 -- només utilitzant join
143 • SELECT iban, ROUND(AVG(amount), 2) AS avg_amount
144 FROM companies c
145 JOIN transactions t ON company_id=business_id
146 JOIN credit_cards cc ON cc.id = card_id
147 WHERE company_name = 'Donec Ltd'
148 GROUP BY iban;

```

100% 25:142

Result Grid Filter Rows: Search Export:

iban	avg_amount
XX77675291784595297555640	257.37
XX413827362289719304908990	139.59
XX347787246070769610780308	240.41
XX688768436543090894854602	188.58
MK28368851538688349	439.39
PL76249283566852676343404576	541.56
XX804008331134918341803913	189.21

Result 26

Action Output

	Time	Response
145	11:16:23	S 371 row(s) returned

Nivell 2

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en *si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu*. Partint d'aquesta taula respon:

- Creació de la taula

```
166 • CREATE TABLE IF NOT EXISTS credit_cards_status(  
167     card_id VARCHAR (255),  
168     status_card VARCHAR (255)  
169 );  
170
```

100% 10:167

	Time	Action	Response
194	12:23:13	CREATE TABLE IF NOT EXISTS credit_cards_status(card_id VARCHAR (255), status_card VARCHAR (255))	0 row(s) affected

- Inserció dades a columna card_id

```
171 -- inserim dades columna card_id  
172 • INSERT INTO credit_cards_status (card_id)  
173     SELECT id  
174     FROM credit_cards;  
175
```

100% 9:173

	Time	Action
195	12:23:34	INSERT INTO credit_cards_status (card_id) SELECT id FROM credit_cards

- Inserció de dades a columna status_card

```
186  
187 • UPDATE credit_cards_status cs  
188     JOIN (SELECT card_id,  
189             CASE WHEN MIN(declined)=0 THEN 'actiu'  
190                  ELSE 'inactiu'  
191             END AS new_status  
192     FROM (SELECT card_id,declined, timestamp,  
193             ROW_NUMBER() OVER(PARTITION BY card_id ORDER BY timestamp DESC) AS row_num  
194     FROM transactions) AS co  
195     WHERE row_num <= 3  
196     GROUP BY card_id) AS card_declined  
197     ON cs.card_id = card_declined.card_id  
198     SET status_card = new_status;  
199
```

100% 16:192

	Time	Action	Response
189	12:15:27	UPDATE credit_cards_status cs JOIN (SELECT card_id, CASE WHEN MIN(declined)=0 THEN 'actiu' ELSE 'inactiu' END AS new_status FROM (SELECT card_id,declined, timestamp, ROW_NUMBER() OVER(PARTITION BY card_id ORDER BY timestamp DESC) AS row_num FROM transactions) AS co WHERE row_num <= 3 GROUP BY card_id) AS card_declined ON cs.card_id = card_declined.card_id SET status_card = new_status;	0 row(s) affected Rows

Exercici 1

Quantes targetes estan actives?

```
201  /*Exercici 1
202  Quantes targetes estan actives? */
203
204  • SELECT COUNT(card_id)
205  FROM credit_cards_status
206  WHERE status_card = 'actiu';
```

100% 16:205

Result Grid

COUNT(card_id)
4995

Result 33 Result 42

Action Output

	Time	Action
✓ 191	12:19:56	SELECT COUNT(card_id) FROM credit_cards_status WHERE status_card = 'actiu'

Nivell 3

Crea una taula amb la qual puguem unir les dades del nou arxiu products.csv amb la base de dades creada, tenint en compte que des de transaction tens product_ids. Genera la següent consulta:

** cross join

- Creació taula products

```
214  -- creacio de la taula
215  • CREATE TABLE IF NOT EXISTS products(
216      id VARCHAR (255) PRIMARY KEY,
217      product_name VARCHAR (255),
218      price VARCHAR(255),
219      colour VARCHAR (255),
220      weight VARCHAR (255),
221      warehouse_id VARCHAR (255)
222  );
223
```

100% 13:219

Action Output

	Time	Action
✓ 199	12:37:22	CREATE TABLE IF NOT EXISTS products(id VARCHAR (255) PRIMARY KEY, product_name VARCHAR (255),

- Carrega de dades products.csv

```

224
225 -- carreguem data de l'arxiu .csv de la taula products
226 • LOAD DATA
227   LOCAL INFILE '/Users/didi/Downloads/products.csv'
228   INTO TABLE products
229   FIELDS TERMINATED BY ','
230   IGNORE 1 ROWS;
231
100% 11:227

```

Action Output		
	Time	Action
✓ 200	12:38:46	LOAD DATA LOCAL INFILE '/Users/didi/Downloads/products.csv' INTO TABLE products FIELDS TERMINATED BY ',' IGNORE 1 ROWS

Per crear la FK a transactions abans hem de desdoblar la columna product_ids de la mateixa i tenir la taula estructurada. L'hem de insertar dins d'una fact-table entre products i transactions.

- Per crear la taula primer canviarem el tipus de data de la id de la taula products a tipus INT així en estalviarem els espais a l'hora de desconcatenar el format json.

```

257 -- canviem les cloumnes a tipus INT de products i de transaction-products
258 • ALTER TABLE products
259   MODIFY id INT;
260
100% 6:258

```

Action Output			
	Time	Action	Response
✓ 26	13:00:22	SELECT * FROM sprint_04_transactions.trans...	253391 row(s) returned
✓ 27	13:03:34	DROP TABLE `sprint_04_transactions`.`trans...	0 row(s) affected
✓ 28	13:03:38	CREATE TABLE IF NOT EXISTS transaction_pr...	0 row(s) affected
✓ 29	13:04:00	INSERT INTO transaction_products (transacti...	253391 row(s) affected Records: 253391 Duplicates: 0
✗ 30	13:06:04	INSERT INTO transaction_products (transacti...	Error Code: 1062. Duplicate entry '00043A49-2949-49
✓ 31	13:06:07	ALTER TABLE products MODIFY id INT	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

- Creació taula transactions_products

```

232 -- creacio taula relacional entre transactions i products
233
234 • CREATE TABLE IF NOT EXISTS transaction_products (
235     transaction_id VARCHAR (255),
236     product_id INT,
237     PRIMARY KEY (transaction_id, product_id),
238     FOREIGN KEY (product_id) REFERENCES products(id),
239     FOREIGN KEY (transaction_id) REFERENCES transactions(id)
240 );
241
100% 17:237

```

	Time	Action	Response
23	12:55:39	CREATE TABLE IF NOT EXISTS transaction_pr...	Error Code: 3780. Referencing column 'product
24	12:59:05	CREATE TABLE IF NOT EXISTS transaction_pr...	0 row(s) affected
25	12:59:09	INSERT INTO transaction_products (transacti...	253391 row(s) affected Records: 253391 Dupli
26	13:00:22	SELECT * FROM sprint_04_transactions.trans...	253391 row(s) returned
27	13:03:34	DROP TABLE `sprint_04_transactions`.`trans...	0 row(s) affected
28	13:03:38	CREATE TABLE IF NOT EXISTS transaction_pr...	0 row(s) affected

- Càrrega de dades a transactions_products

```

242 -- carrega de dades
243 • INSERT INTO transaction_products (transaction_id, product_id)
244     SELECT t.id, jt.product_id
245     FROM transactions AS t
246     JOIN JSON_TABLE(
247         CONCAT('[', product_ids, ']'),
248         "$[*]" COLUMNS (product_id INT PATH "$")
249     ) AS jt;
250
100% 11:245

```

	Time	Action	Response
23	12:55:39	CREATE TABLE IF NOT EXISTS transaction_pr...	Error Code: 3780. Referencing column 'prc
24	12:59:05	CREATE TABLE IF NOT EXISTS transaction_pr...	0 row(s) affected
25	12:59:09	INSERT INTO transaction_products (transacti...	253391 row(s) affected Records: 253391
26	13:00:22	SELECT * FROM sprint_04_transactions.trans...	253391 row(s) returned
27	13:03:34	DROP TABLE `sprint_04_transactions`.`trans...	0 row(s) affected
28	13:03:38	CREATE TABLE IF NOT EXISTS transaction_pr...	0 row(s) affected
29	13:04:00	INSERT INTO transaction_products (transacti...	253391 row(s) affected Records: 253391

- La taula és la següent

1 • `SELECT * FROM sprint_04_transactions.transaction_products;`

100% 1:1

Result Grid

Filter Rows:

Edit:   

Export/Import:  

Fetch rows: 

transaction_id	product_...
00043A49-2949-494B-A5DD-A5BAE3BB19DD	97
00043A49-2949-494B-A5DD-A5BAE3BB19DD	16
000447FE-B650-4DCF-85DE-C7ED0EE1CAAD	69
000447FE-B650-4DCF-85DE-C7ED0EE1CAAD	87
000447FE-B650-4DCF-85DE-C7ED0EE1CAAD	66
00045D6B-ED2E-4F2F-8186-CEE074D875D0	11
00045D6B-ED2E-4F2F-8186-CEE074D875D0	16
00045D6B-ED2E-4F2F-8186-CEE074D875D0	81
00045D6B-ED2E-4F2F-8186-CEE074D875D0	20

transaction_products 1

Action Output 

	Time	Action
✓ 209	13:18:35	SELECT * FROM sprint_04_transactions.transaction_products

Exercici 1

Necessitem conèixer el nombre de vegades que s'ha venut cada producte.

275 ↳ Necessitem conèixer el nombre de vegades que s'ha venut cada producte. */

276

277 •

```
SELECT product_id, COUNT(product_id) AS sales_for_product
```

278

```
FROM transaction_products
```

100% 58:277

Result Grid

Filter Rows: Search

Export:

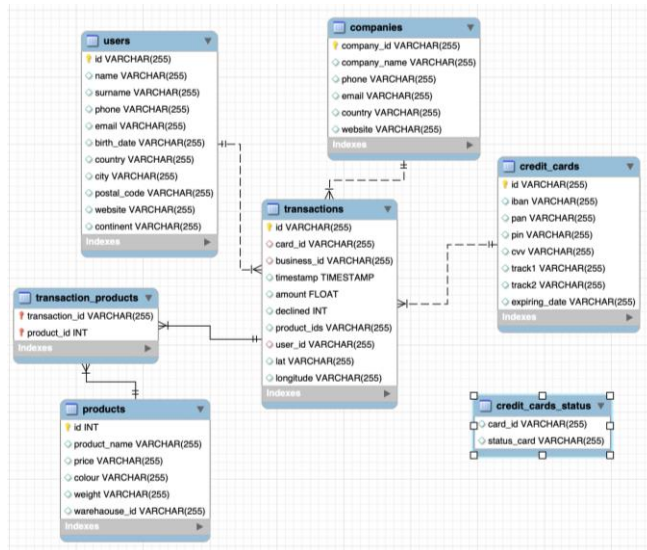
product_id	sales_for_product
22	2448
19	2449
69	2457
87	2461

Result 5

Action Output

	Time	Action	Response
✓ 29	13:04:00	INSERT INTO transaction_products (transacti...	253391 row(s) affected Records: 253391 Duplicates: 0
✗ 30	13:06:04	INSERT INTO transaction_products (transacti...	Error Code: 1062. Duplicate entry '00043A49-2949-494B-A5DD-A5BAE3BB19DD' for key 'PRIMARY'
✓ 31	13:06:07	ALTER TABLE products MODIFY id INT	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 32	13:07:39	SELECT TABLE_NAME, CONSTRAINT_...	4 row(s) returned
✓ 33	13:16:43	SELECT product_id, COUNT(product_id) FRO...	100 row(s) returned
✓ 34	13:18:11	SELECT product_id, COUNT(product_id) AS s...	100 row(s) returned

DIAGRAMA FINAL



La taula seleccionada “credit_cards_status” es podria haver substituït per una vista per reduir pes i cost.

La taula no està relacionada amb les altres.